NEPHROLOGY TEACHING SERIES

Glomerulonephritis

Active sediment, pattern recognition, urgent biopsy

Hematuria + proteinuria + rising Cr is not just AKI — look at the urine.

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MS3/MS4 rotation teaching

Why we get consulted

Rapidly progressive GN can cause irreversible loss if not recognized early.

TRIGGERS FOR CONSULT

- Active sediment: RBC casts, dysmorphic RBCs
- Hematuria + proteinuria + rising Cr
- Nephrotic range (UPCR > 3 or 3.5 g/day)
- Pulmonary-renal syndrome
- Rash + arthritis + kidney disease
- Unexplained AKI with active UA
- Hypocomplementemia + AKI

TEACHING PEARL

Hematuria + proteinuria + rising Cr is never 'just AKI.' Urine findings can flip management from watchful ATN to urgent biopsy + immunosuppression.

KDIGO 2021 Clinical Practice Guideline for Glomerular Diseases.

Nephritic vs nephrotic — the orienting split

Two patterns, different differentials, different urgencies.

NEPHRITIC — inflammation

Hematuria (dysmorphic RBCs, RBC casts)

Modest proteinuria (usually < 3 g/day)

Hypertension (salt + water retention)

Rising Cr, often acute/subacute

Edema — periorbital, pedal

Think: IgA, post-infectious, lupus, antineutrophil cytoplasmic antibody (ANCA), anti-glomerular basement membrane (anti-GBM)

Urgency: rising Cr = consider rapidly progressive glomerulonephritis (RPGN) workup

NEPHROTIC — barrier loss

Proteinuria > 3.5 g/day (or UPCR > 3)

Hypoalbuminemia (< 3 g/dL)

Edema — often generalized, anasarca

Hyperlipidemia, hypercoagulability (renal vein thrombosis, DVT)

Bland sediment (± oval fat bodies, 'maltese cross')

Think: minimal change, focal segmental glomerulosclerosis (FSGS), membranous, DKD, amyloid

Urgency: usually less acute, but assess VTE risk, edema, and lipids

UpToDate: Overview of the classification and causes of glomerular disease (2026).

Proteinuria and nephrotic syndrome

Quantify first — then recognize the complications.

QUANTIFY PROTEINURIA

UACR — best screen (albumin specific)

Normal < 30 mg/g

Moderate A2 = 30–300

Severe A3 = > 300

UPCR (g/g) ≈ g/day proteinuria

Nephrotic range: UPCR > 3 or > 3.5 g/day

24-h urine: gold standard but often impractical

NEPHROTIC SYNDROME — MANAGE

Tetrad: > 3.5 g/day proteinuria, albumin < 3, edema, hyperlipidemia

VTE risk — consider anticoagulation when albumin is very low, especially membranous, after bleeding-risk review

Infection risk (loss of Ig, complement)

Tx: max ACEi/ARB, SGLT2i, statin, Na restriction

Biopsy confirms primary cause — drives immunosuppression

KDIGO 2021 Glomerular Diseases. UpToDate: Overview of nephrotic syndrome (2026).

UA sediment — reference card

Optional reference. Most labs report sediment findings automatically.

Finding	What it means
RBC casts / dysmorphic RBCs	Glomerular bleeding → GN (IgA, ANCA, lupus, anti-GBM, post-strep)
Muddy brown / granular casts	Tubular injury → ATN (ischemic, septic, toxic)
WBC casts	Pyelonephritis, acute interstitial nephritis (AIN)
Waxy casts	Chronic kidney disease (advanced, low urine flow)
Hyaline casts	Nonspecific — dehydration, fever; can be normal
Oval fat bodies / Maltese cross	Heavy proteinuria / nephrotic syndrome
Eosinophils	AIN (low sensitivity) or atheroembolic disease
Crystals (calcium oxalate, uric acid)	Ethylene glycol (oxalate); tumor lysis / urate; stones

***Note:** most outpatient labs report sediment automatically; manual microscopy is rarely done in clinic but remains the gold standard when the diagnosis is unclear.*

UpToDate: Urinalysis in kidney disease (2026).

The GN serologic panel

Send it early — results often take days, and timing matters.

Test	What it tells you	Disease
ANA, anti-dsDNA, anti-Sm	Lupus activity; dsDNA correlates with nephritis	SLE nephritis
C3 / C4	Low in SLE, post-strep, membranoproliferative glomerulonephritis (MPGN), cryoglobulinemia; normal in ANCA and anti-GBM	Narrow ddx quickly
ANCA (myeloperoxidase (MPO), proteinase 3 (PR3))	PR3 → granulomatosis with polyangiitis (GPA); MPO → microscopic polyangiitis (MPA); either can cause renal-limited vasculitis	ANCA-associated vasculitis
Anti-GBM	Linear IgG on biopsy; rapid and irreversible	Goodpasture syndrome
PLA2R antibody	Positive in ~70% of primary membranous; follow to monitor therapy	Membranous nephropathy
HBV, HCV, HIV, cryoglobulins	Secondary causes of MPGN or membranous	Infection-driven GN
Complement CH50	Screen for C3 glomerulopathy / dense deposit disease	Rare complement-mediated GN
ASO, anti-DNase B	Recent strep infection (2–4 weeks)	Post-streptococcal GN

UpToDate: Evaluation of the adult with hematuria (2026).

Urgent: pulmonary-renal syndrome & RPGN

Minutes-to-hours decisions. Biopsy and immunosuppression don't wait.

RED FLAGS

- Hemoptysis + hematuria + rising Cr
- New oxygen requirement, infiltrates on CXR
- Cr doubling over days; oliguria
- Active sediment (RBC casts)
- Mononeuritis multiplex, sinus disease, upper-airway bleeding
- ANCA or anti-GBM positive
- **Action: biopsy TODAY if feasible; empiric steroids + consider PLEX**

INITIAL MANAGEMENT (before biopsy)

- **Stabilize: O2, transfuse if bleeding, protect access**
- Send: ANCA, anti-GBM, ANA, dsDNA, C3/C4, urgent UA + microscopy
- Empiric IV methylprednisolone 500–1000 mg/day × 3 if anti-GBM suspected (can't wait)
- PLEX: consider for severe kidney disease, rapidly rising Cr, dialysis need, or hypoxemic pulmonary hemorrhage
- Biopsy as soon as platelets, BP, anticoagulation allow
- Consult IR / nephrology for urgent native kidney biopsy

UpToDate: Approach to the patient with pulmonary-renal syndrome (2026).

Common glomerular diseases — recognition

Pattern + clinical context = diagnosis most of the time.

Disease	Presentation	Clue
IgA nephropathy	Recurrent synpharyngitic hematuria in young adult	IgA-dominant IF on biopsy; most common GN worldwide
Post-infectious GN	Kids 1–3 wk after strep throat/impetigo; adults post-staph	Low C3 (recovers 6–8 wk); humps on EM
Lupus nephritis	Young woman, rash, arthritis, serositis, cytopenias	ANA+, dsDNA+, low C3/C4; class I–VI on biopsy
ANCA vasculitis (GPA, MPA)	Older adult, sinus/pulmonary disease, RPGN	MPO/PR3+; pauci-immune crescentic GN
Anti-GBM (Goodpasture)	Hemoptysis + RPGN, bimodal age	Linear IgG; urgent PLEX + immunosuppression
Membranous nephropathy	Adult nephrotic syndrome + VTE	PLA2R+ in ~70% (primary); spike & dome on EM
FSGS	African ancestry/APOL1 risk, obesity, HIV, heroin, reflux — nephrotic range proteinuria	Segmental sclerosis; APOL1 variants
Minimal change	Sudden full nephrotic syndrome; children; NSAID-associated in adults	Normal LM, foot-process effacement on EM

UpToDate: various topic reviews (2026).

Kidney biopsy + treatment scaffolding

Biopsy confirms; supportive care + targeted immunosuppression treats.

WHEN TO BIOPSY

- Rapidly rising Cr + active sediment
- Nephrotic syndrome in adult (most — exclude DKD)
- Unexplained AKI with hematuria/proteinuria
- Suspected lupus nephritis to classify
- Isolated persistent proteinuria > 1 g/day with features
- **Pre-biopsy: platelets > 50K, INR < 1.5, BP < 160/90, stop AC 5–7 d**
- Complications: bleeding (~2%), transfusion (~1%), AV fistula, rarely nephrectomy

SUPPORTIVE THERAPY (always)

- Max-dose ACEi or ARB for proteinuria
- SGLT2i (added benefit in proteinuric CKD — DAPA-CKD, EMPA-KIDNEY)
- BP target SBP < 120 (proteinuric, non-dialysis CKD)
- Low-sodium diet (< 2 g/day)
- Statin for lipid management (nephrotic)
- Consider anticoagulation for high-risk nephrotic syndrome after bleeding-risk review
- Vaccinations before immunosuppression (PPSV, HBV, influenza, COVID, zoster)

KDIGO 2021 Glomerular Diseases Guideline.

Landmark trials in glomerular disease

The RCTs that shape current management.

Trial	Year	Takeaway
RAVE	2010	Rituximab non-inferior to cyclophosphamide for ANCA induction; preferred in relapse.
MAINRITSAN	2014	Rituximab maintenance superior to azathioprine in ANCA vasculitis.
PEXIVAS	2020	Reduced-dose steroids non-inferior; PLEX not routine, but KDIGO 2024 still considers it in highest-risk AAV.
MENTOR	2019	Rituximab SUPERIOR to cyclosporine at 24 mo for membranous (durable remission).
STOP-IgAN	2015	Immunosuppression on top of supportive care did NOT slow IgAN; SC is foundation.
TESTING	2022	Methylpred slowed IgAN progression (HR 0.53 full cohort; HR 0.27 reduced-dose); more infection.
NefIgArd	2023	Targeted-release budesonide ↓ proteinuria & slowed eGFR decline in IgAN.
DAPA-CKD	2020	Dapagliflozin ↓ kidney failure even in non-DM CKD (including IgAN, FSGS subgroups).
EMPA-KIDNEY	2023	Empagliflozin benefit across eGFR/urine albumin-to-creatinine ratio (UACR), including many primary GN etiologies.

Full citations on references slide.

Common presentation mistakes

Avoid these when presenting a possible GN.

AVOID

- Calling it AKI without checking UA
- Relying on dipstick alone
- Missing day-1 serologic panel
- Delaying biopsy for labs
- Steroids without a taper plan
- Automatic anticoagulation for every nephrotic patient
- No vaccines before immunosuppression

DO INSTEAD

- Lead with UA + proteinuria
- Request formal UA + microscopy
- Day 1: ANA, dsDNA, ANCA, anti-GBM, C3/C4
- Tee up biopsy early (plts, INR, BP)
- Pair steroids with rituximab/CYC + taper
- Low albumin → weigh VTE vs bleeding risk
- Vaccinate before steroids

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 48-year-old man, 3-week history of sinusitis, hemoptysis, and fatigue. Admitted with Cr 3.4 (baseline 1.0 six months ago), UA: 3+ blood, 2+ protein, RBC casts, dysmorphic RBCs; UACR 2800. BP 168/102. CXR: bilateral patchy infiltrates. C3 and C4 normal. ANCA pending. No rash.

QUESTION: What's the likely diagnosis, what do you send today, and what's the urgency?

Clinical case — answer

Key teaching points from the case.

ANSWER

ANCA-associated vasculitis (likely GPA: sinus + pulmonary + renal, normal complements). Send ANCA (MPO/PR3), anti-GBM, ANA/dsDNA. Biopsy TODAY. Empiric high-dose IV methylprednisolone after samples drawn.

TEACHING

Pulmonary-renal syndrome with rapidly rising Cr + active sediment = RPGN until proven otherwise. Classic triad for GPA: upper airway (sinus, nose) + lower airway (lung) + kidney. Normal C3/C4 argues against lupus or post-infectious. While waiting for biopsy: stabilize BP, oxygenation, no anticoagulation if biopsy planned. RAVE showed rituximab non-inferior to cyclophosphamide; PEXIVAS supports reduced-dose steroids. PLEX is not routine, but KDIGO 2024 still considers it for SCr > 3.4 mg/dL, dialysis/rapidly rising SCr, hypoxemic alveolar hemorrhage, or ANCA/anti-GBM overlap.

REFERENCES

Glomerulonephritis

Guidelines

KDIGO 2021 Glomerular Diseases Guideline
KDIGO 2024 ANCA Vasculitis Guideline
ACR 2019 Lupus Nephritis Guidelines
EULAR/ERA-EDTA SLE Management Recs
CHAMP — ANCA vasculitis management consensus
ASN/NephJC educational resources

Landmark Trials

RAVE — NEJM 2010
MAINRITSAN — NEJM 2014
PEXIVAS — NEJM 2020
MENTOR — NEJM 2019
STOP-IgAN — NEJM 2015
TESTING — JAMA 2022
NefIgArd — Lancet 2023
DAPA-CKD — NEJM 2020
EMPA-KIDNEY — NEJM 2023

UpToDate (2026 access):

Overview of the classification and causes of glomerular disease • Evaluation of the adult with hematuria • Approach to the patient with pulmonary-renal syndrome • Treatment of ANCA-associated vasculitis • Treatment of lupus nephritis • Primary membranous nephropathy: pathogenesis and treatment